

DAY 171/365 • PART 2 OF 3

How does a tree pick its questions?

Entropy, information gain and Gini, in plain words.

Many questions. Which one first?

For a customer dataset, the tree could ask about age, income, city or past purchases. Asking in the wrong order gives a messy tree.

Rule of thumb: ask the question that leaves the groups as pure as possible.

To compare questions, we need a score for how mixed a group is. Take three bags of balls:

Bag	Mix	How messy?
A	5 red, 0 blue	Not at all
B	4 red, 1 blue	A little
C	5 red, 5 blue	Maximum

Two popular scores do this job: **entropy** and **Gini**.

Entropy: how uncertain is the group?

Entropy is the uncertainty in a group. If you pick one item at random, how hard is it to guess its label?

Group	Entropy
All "Yes" (5 : 0)	0, no doubt
Mostly "Yes" (4 : 1)	0.72
3 buy, 5 don't (3 : 5)	0.95
Even split (5 : 5)	1.0, a coin flip

Lower is better. 0 means pure. For two classes, the maximum is 1.

Formula: $-\sum p \times \log_2(p)$, where p is the share of each class.

Information gain: how much did the question help?

Information gain

= entropy before the split

– average entropy after the split

The average is **weighted**, so a big group counts more than a tiny one.

In other words: how much did the uncertainty drop?

Big drop

Young people all bought,
older people all didn't.
Gain is high.

No drop

Both groups are as mixed
as before. Gain is about
0.

The tree picks the question with the highest information gain.

Worked example: 10 customers

5 bought and 5 didn't, so the starting entropy is 1.0. We test three questions:

Question	Groups after split	Gain
Bought before?	5 : 0 and 0 : 5, both pure	1.00
Age > 30?	4 : 1 and 1 : 4	0.28
Likes blue?	3 : 2 and 2 : 3	0.03

"Bought before?" wins and becomes the root.
"Likes blue?" barely helps, so the tree ignores it.

Then the tree repeats this on each child group that's still mixed.

Illustrative numbers, with gains calculated from the entropy formula.

Gini index: the other way to measure mess

It asks: if I pick an item at random and label it randomly using the group's mix, how often would I be wrong?

Formula: $1 - \sum p^2$

Group	Gini	Entropy
5 : 0	0	0
4 : 1	0.32	0.72
5 : 5	0.50	1.00

Same idea: lower is better, 0 is pure. For two classes, the maximum Gini is 0.5. In scikit-learn, "gini" is the default.

Gini or entropy?

Gini

Faster to compute. It can favour splits that make equal-sized groups.

Entropy

Pairs with information gain. Slightly slower because of the log.

In practice they often pick very similar trees, as the table on the last slide showed: the two scores rank the groups the same way.

My plan: treat it like a setting. Try both and let cross-validation decide, the same way I tuned alpha in my Ridge series.

TAKEAWAYS

A tree asks the question that leaves the purest groups.

Entropy and Gini measure how mixed a group is. Lower is purer.

Information gain is the drop in entropy after a split. The highest gain wins.

Part 3 tomorrow: a full worked tree on a tiny dataset, plus how to stop it from overfitting.